

Owner / Executive Role Specification

Comprehensive Feature Catalog, Business Value Rationale, Data Contracts & Backend API Specifications.

Role Identifier: `admin` (Company Owner / GC Principal) **Persona:** Avery Scott (Managing Partner)

Platform: LATTICE Construction Management **Version:** 1.0.0 (Production Ready)

1. Executive Overview & Mental Model

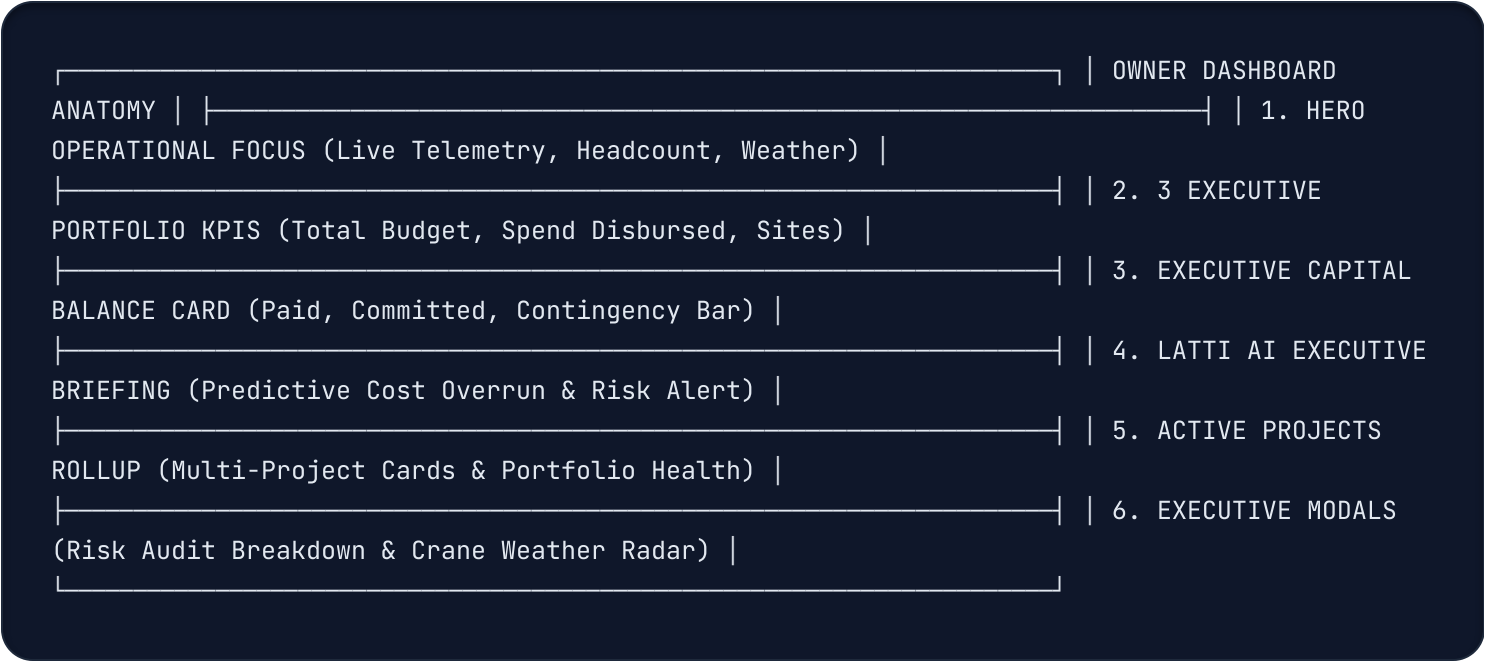
The **Company Owner** (Avery Scott) operates at the 30,000-foot strategic layer of the construction enterprise. Unlike a Field Superintendent managing daily trade crews or a Project Manager handling RFIs and submittals, **the Owner cares about portfolio solvency, bank draw status, risk containment, and overall capital protection.**

Core Executive Questions Answered by This Dashboard:

- Is our overall portfolio capital (\$34.85M) healthy and protected against margin erosion?
- Are active site variances being absorbed by contingency reserves without dipping into company profits?
- Did trade crews show up today (Live Headcount: 94 on site)?
- Are there critical-path inspection roadblocks that threaten client handover dates?

2. Visual Architecture & Layout Anatomy

The Owner dashboard is structured into 5 cohesive tiers designed for 3-second rapid comprehension:



3. Comprehensive Feature-by-Feature Breakdown

Feature 1: Hero Operational Focus & Site Telemetry Card

Visual Presentation: Soft gradient card (#EAF3FF to #FFFFFF) with blue hairline border.

Business Value: Provides an immediate visual pulse of today's highest-stakes operational milestone across all 7 projects.

- **Today's Focus Badge:** Highlights the priority event of the day.
- **Live Headcount (94 on site):** Pulsing green dot showing aggregated manpower across active jobsites. If manpower drops below threshold, owners know delays are imminent.
- **Milestone Headline (City Framing Walkthrough):** Highlights the critical-path inspection with City of Tampa Inspector arriving at 10:00 AM.
- **Live Weather Pill (82°F Sunny):** Tapping opens the 3-day crane radar and concrete cure risk modal.

Backend API Endpoint: GET /api/v1/owner/site-pulse

```
{
  "focusMilestone": {
    "id": "ms-101",
    "projectId": "proj-1",
    "projectName": "Snell Isle Residence",
    "title": "City Framing Walkthrough",
    "scheduledTime": "2026-09-05T10:00:00Z",
    "inspector": "Frank Rodriguez (City of Tampa)",
    "criticalPath": true
  },
  "liveTelemetry": {
    "totalHeadcountOnSite": 94,
    "activeSubcontractorCrews": 6,
    "timestamp": "2026-09-05T08:30:00Z"
  },
  "weather": {
    "tempFahrenheit": 82,
    "condition": "Sunny",
    "windSpeedMph": 8,
    "precipitationChance": 10,
    "isCraneHalted": false
  }
}
```

Feature 2: The 3 Core Executive KPIs

Positioned directly below the hero card in a clean 3-column responsive grid.

Metric Card	Value	Executive Meaning & Rationale	Backend Aggregation Formula
Total Budget	\$34.85M	Sum of all signed client contracts under active construction.	SUM(projects.budget.total)
Total Spend	\$16.82M	Cumulative capital paid out to trades and vendors via ACH/wire.	SUM(projects.budget.actual + projects.budget.paid)
Active Projects	7 Active	Count of active jobsites under construction.	COUNT(projects WHERE status = 'In Progress')

Feature 3: Executive Capital Balance & Contingency Reserve Card

The Problem: Construction projects fail when unexpected site conditions eat through contingency buffers, destroying company margins.

The Solution: Displays real-time breakdown of total company capital into 3 vital buckets:

- **PAID 48% (\$16.8M)** — Capital already wired to trades.
- **COMMITTED 35% (\$12.1M)** — Signed subcontracts and purchase orders pending billing.
- **CONTINGENCY 17% (\$5.95M)** — Unspent buffer protecting company profit margins.
- **Status Badge (**PROTECTED**):** Reassures the owner that cost variances have not breached the reserve threshold.

Feature 4: Latti AI Executive Briefing (Predictive Risk Intelligence)

Synthesizes 500+ cost line items into a crisp, 2-sentence executive summary:

Real-World Scenario Shown:
"Snell Isle concrete costs are running 8% (+\$14,200) over budget due to revised pier depths. Thursday rainfall threatens exterior concrete cure. All other projects are tracking within contingency thresholds."

- **Button 1: **Inspect Budget Risks**** : Opens the Executive Risk Audit modal detailing the CSI 03 30 00 soil test variance.
- **Button 2: **Ask Copilot**** : Opens Latti AI chat preloaded with *"Show owner risk analysis"*.

Feature 5 & 6: Multi-Project Rollup & Executive Modals

- **Active Projects Rollup:** Shows individual cards for Snell Isle, Downtown Commercial, Greenfield, and Sunset Villas with thumbnail, contract value, schedule status, and PM contact.
- **Executive Risk Audit Modal:** Details geotechnical soil findings from Madrid CPWG requiring piers deepened from 18ft to 24ft, confirming that Snell Isle's \$185K Phase 1 contingency absorbed the cost with zero client delay.
- **Weather Impact Modal:** Shows 3-day radar, high-wind crane stop threshold (>25 mph), and schedule adjustments.

| 4. Recommended Database Schema (PostgreSQL / Relational)

```
-- 1. Projects Master Table
CREATE TABLE projects (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name VARCHAR(255) NOT NULL,
  code VARCHAR(50) UNIQUE NOT NULL,
  city VARCHAR(100),
  state VARCHAR(50),
  status VARCHAR(50) NOT NULL DEFAULT 'In Progress',
  start_date DATE NOT NULL,
  target_end_date DATE NOT NULL,
  progress_percentage NUMERIC(5,2) DEFAULT 0.00
);

-- 2. Project Budgets & Contingencies
CREATE TABLE project_budgets (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  project_id UUID REFERENCES projects(id) ON DELETE CASCADE,
  total_budget NUMERIC(15,2) NOT NULL,
  committed_amount NUMERIC(15,2) DEFAULT 0.00,
  actual_spent_amount NUMERIC(15,2) DEFAULT 0.00,
  contingency_reserve NUMERIC(15,2) DEFAULT 0.00,
  currency VARCHAR(3) DEFAULT 'USD'
);

-- 3. CSI Cost Code Variances Table
CREATE TABLE cost_code_variances (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  project_id UUID REFERENCES projects(id) ON DELETE CASCADE,
  csi_code VARCHAR(20) NOT NULL, -- e.g. '03 30 00'
  division_name VARCHAR(100) NOT NULL, -- 'Cast-in-Place Concrete'
  budgeted_amount NUMERIC(15,2) NOT NULL,
  actual_amount NUMERIC(15,2) NOT NULL,
  variance_amount NUMERIC(15,2) GENERATED ALWAYS AS (actual_amount - budgeted_amount) STORED,
  root_cause_summary TEXT,
  is_contingency_absorbed BOOLEAN DEFAULT TRUE
);
```

5. Backend টিমকে যেভাবে সংক্ষেপে উপস্থাপন করবেন (Quick Presentation Guide in Bengali)

ডেভেলপারদের সাথে আলোচনার মূল বিষয়গুলো:

1. **অনারের মূল চাহিদা:** অনার কোনো টুলস বা চেকলিস্ট দেখতে চান না। তিনি দেখেন মোট কত টাকা ইনভেস্ট হয়েছে (*Total Budget*), কত খরচ হয়েছে (*Total Spend*), এবং কোম্পানির লাভের জন্য রাখা ইমার্জেন্সি বাফার (*Contingency*) ঠিক আছে কিনা।
2. **লাইভ হেডকাউন্ট (94 on site):** ফিল্ড সুপারিনটেনডেন্ট সকালে যে ডেইলি লগ তৈরি করেন, সেখানকার সব সাবকন্ট্রাক্টরের লোকবল যোগ করে স্বয়ংক্রিয়ভাবে অনার ড্যাশবোর্ডে লাইভ দেখানো হয়।
3. **ক্যাপিটাল ব্যালেন্স বার (Paid 48%, Committed 35%, Contingency 17%):** ব্যাকএন্ডকে এই ৩টি ফিল্ডের রেশিও এপিআই-এর মাধ্যমে পাঠাতে হবে।
4. **Latti AI ও রিস্ক অডিট:** যখনই কোনো কস্ট কোডে বাজেট অতিরিক্ত খরচ হয় (যেমন: ফাউন্ডেশন পাইলিংয়ে +\$14,200), তখন ব্যাকএন্ড AI ইঞ্জিন এটি ডিটেক্ট করে অনারকে জানাবে যে টাকাটি কন্টিনজেন্সি রিজার্ভ থেকে কভার হয়েছে কিনা।